

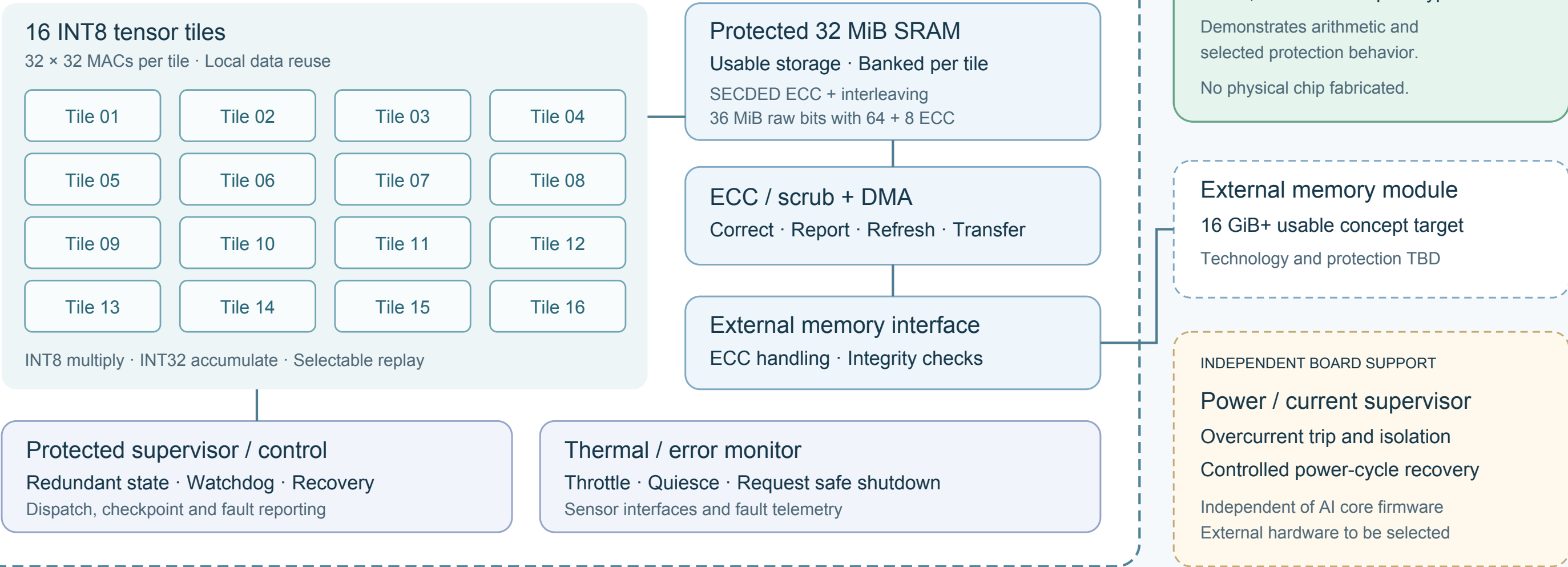
ORBIT-AI / First-pass chip concept

CONCEPT / v0.1

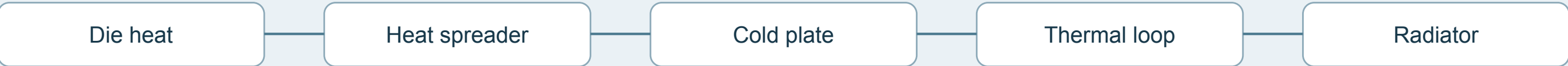
Space-based AI computing platform · Inference first · Training extension planned

Proposed target architecture

Conceptual blocks - not a physical layout



PROPOSED MODULE / SPACECRAFT THERMAL PATH



Conduction into the thermal system; radiator sizing depends on orbit, attitude and environmental heat loads.

6.55-26.2 dense INT8 TOPS at 200-800 MHz target

40 W chip sizing assumption - not an estimate

All proposed capacity and performance labels are unvalidated targets. No radiation qualification, physical timing or thermal signoff is claimed.

Architecture and AI performance

Inference first; training is a separate, future capability.

The proposed chip is a recoverable compute building block for an orbital AI platform. The delivered hardware-description prototype is deliberately much smaller. Mission orbit, lifetime, manufacturing process, package and radiation acceptance levels remain undecided.

Element	First-pass target or assumption
AI engine	16 tiles x 32 x 32 INT8 multiply-accumulate units; INT32 accumulation.
Clock and peak	Explore 200-800 MHz: 6.55-26.21 dense INT8 TOPS, counting a MAC as two operations. No timing measurement supports these clock targets.
Local memory	32 MiB usable, distributed 2 MiB per tile. 64+8 SECEDED expands storage to 36 MiB raw bits, before spares and periphery.
External memory	16 GiB+ usable capacity; 256 GB/s usable bandwidth scenario. Memory technology and qualified parts are unselected.
Power allocation	40 W per chip for conceptual sizing only. This is not a power estimate and excludes external memory and module overhead.
Protected execution	Proposed selectable duplication, compare, checkpoint/retry and tile quarantine. Fully duplicated work halves useful peak throughput.
Training extension	Proposed BF16/FP16 multiplication, FP32 accumulation/reduction/update paths and training software. None is in the delivered INT8 prototype.

Maximize useful AI work, not just arithmetic count

Banked local memory, double buffering, retained partial sums and weight/activation reuse are the main priorities [9,10]. At 800 MHz, a conventional 32 x 32 array may consume 51.2 GB/s of local input per tile, or 819.2 GB/s across 16 tiles. This is internal bandwidth; data reuse must reduce external traffic.

Memory-bound example: at 256 GB/s, a streaming INT8 matrix-vector workload near two operations per weight byte has an ideal ceiling of only **0.512 TOPS**. Batched matrix operations can reuse weights. The 26.21 TOPS peak requires at least 102.4 operations per external byte, before other inefficiencies [9].

All performance figures above are calculated bounds. No workload benchmark, utilization, power, area or throughput-per-watt measurement has been made. Training needs additional numerical and software support [11].

Radiation protection and cooling

Layered fault handling plus a real spacecraft heat-rejection path.

Radiation: process, circuits and system must work together

Process and layout: select radiation-characterized cells and memory macros; evaluate total ionizing dose, displacement damage, single-event effects and latch-up under mission conditions. Separate redundant state physically and protect clock/reset distribution. RTL redundancy cannot establish device radiation hardness [4,5,7].

Digital architecture: triplicate critical control, protect memories with ECC, interleave bits, scrub memory and report error rates. Add checkpoint/retry and quarantine faulty tiles. Shared voters, comparators, inputs, clocks and correlated faults need their own analysis. Shield thickness and material require environment and geometry modeling [5,8].

Independent module supervision: current sensing, current limiting, power-domain isolation, watchdog and controlled power cycling. Some destructive events cannot be prevented merely by detecting them and switching power [5,7].

Thermal: carry heat from silicon to an exposed radiator

Die -> spreader -> cold plate -> heat pipe or pumped loop -> radiator. In vacuum, onboard fans cannot reject waste heat to space. The package/loop must conduct heat away and the radiator must radiate it. Orbit and attitude determine external heat loads [6].

Prototype thresholds are illustrative: normal operation below the throttle trigger, alternate-cycle admission from 80 C, stop at 95 C or invalid sensor data, and recovery at 70 C. These are not device ratings. Physical sensors, calibration and an independent shutdown circuit are not implemented.

Radiator temperature	Ideal emitting area / kW	Area / 100 kW
300 K	2.42 m2	242 m2
320 K	1.87 m2	187 m2
350 K	1.31 m2	131 m2

Calculated from $P = \text{emissivity} \times \text{Stefan-Boltzmann constant} \times \text{area} \times T^4$, with emissivity 0.9 and an ideal 0 K view. This is emitting surface area, not necessarily panel footprint. It excludes sunlight, Earth IR/albedo, obstruction, temperature gradients, aging and engineering margin.

Temperature tradeoff: at an assumed 0.5 K/W die-to-radiator resistance and 40 W chip dissipation, the junction is 20 C warmer than the radiator. A 320 K radiator implies 66.85 C; 350 K implies 96.85 C, above the example stop threshold. These lumped steady-state examples are not thermal simulations.



What was built and checked

Yosys 0.69 through YoWASP; generic digital implementation only.

The production Verilog implements four useful signed INT8 MAC lanes, INT32 accumulation, a one-result output buffer, duplicate accumulator/result storage, mismatch fault-stop and a three-copy thermal state with feedback repair. There is no automatic replay, ECC SRAM or floating-point training engine.

Verification	Recorded result
Synthesis / structural check	PASS. 4,356 generic logic cells and 521 flip-flop bits. No PDK mapping or physical area estimate.
Redundancy retention	PASS. Nine storage groups audited; all three thermal state copies and four pairs each of accumulators and output registers retain distinct flip-flop bits.
Arithmetic / handshake / thermal	PASS. Fixed directed scenario through 24 bounded SAT time steps. Includes signed edge cases, output backpressure, throttle, stop and recovery.
Fault injection	PASS. Separate accumulator upset, result-register upset and one thermal-copy upset/repair scenarios, each checked through 8 time steps.
INT32 wraparound	PASS. Boundary scenario through 6 time steps. Overflow wraps; no saturation or overflow alarm.

The synthesis flow explicitly preserves thermal-register drivers; signal attributes alone initially allowed copies to merge. The final netlist audit checks this. Tests inject faults only into generated test copies, and do not add fault ports to the production design.

Coverage limits: these are five bounded, fixed scenarios, not exhaustive correctness proofs or a quantified radiation campaign. Combinational logic can be shared. Voters, comparators, clock/reset, sensor interfaces and handshake flags are not fully protected. A common fault affecting both arithmetic paths may remain undetected.

Next engineering gates

1. Select orbit/lifetime, benchmark models, reliability targets and the module power/cooling envelope.
2. Choose characterized process/library, memory, package and interfaces; implement the full datapath, memory system, supervisor and software.
3. Expand verification and fault injection, map to a real library, place/route, and close timing, power integrity, physical and thermal checks.
4. Fabricate prototypes; perform radiation, thermal-vacuum and mission-specific qualification before deployment.

Project map: RTL and testbenches in **rtl/**; netlist, synthesis log, storage audit and sizing arithmetic in **reports/**; reproducible commands in **README.md** and **rtl/README.md**. Tool versions are pinned in **requirements-tools.txt**.

Sources and assumptions

Primary references consulted on 29 September 2026.

[1] Yosys documentation

Digital synthesis tooling.

[2] YoWASP

Portable tool distribution, including macOS.

[3] OpenROAD platform integration

Requirements for technology-specific physical implementation. OpenROAD was not run.

[4] ECSS-Q-ST-60-15C Rev.1, March 2025

Mission-specific radiation hardness assurance.

[5] ECSS-E-HB-20-40A, October 2023

ASIC/FPGA engineering and radiation mitigation guidance.

[6] NASA: thermal control

Spacecraft conduction, heat transport and radiation.

[7] NASA NESC: mitigating single-event risks

Effects-specific radiation-risk assessment.

[8] NASA: structures, materials and mechanisms

Radiation shielding and environment-dependent modeling.

[9] Berkeley: Roofline model

Arithmetic-intensity and bandwidth limits on performance.

[10] MIT: Eyeriss, ISCA 2016

Dataflow, local reuse and accelerator data-movement energy.

[11] NVIDIA: mixed-precision training

Precision and software requirements for training.

Reading the numbers

The tile count, clock range, memory capacity/bandwidth, 40 W allocation, thermal resistance and temperature thresholds are project assumptions. They are not claims from the references. The arithmetic model in `model/concept_budget.py` documents their consequences. No die dimensions, process node, shield thickness or radiation rating has been invented.